N Satheesh Kumar

Experience 1.5 Years

Software Engineer I, Mirafra Software Technologies – Hyderabad — Mar 2026 – Present
Trainee Software Developer, Hoags Technologies — May 2025 – Jul 2025
Embedded Trainee, Vector India — Mar 2024 – Nov 2024

Skills

- **Programming Languages:** C, Embedded C, Data Structures, C++, Python.
- **Architectures:** Good understanding of ARM, RISC-V, x86.
- **Operating Systems:** Linux, Windows.
- **Microcontrollers / SoCs:** LPC2129 (ARM7 TDMI-S core), Raspberry Pi 4 (BCM2711).
- **Bus Protocols:** UART, I2C, SPI, USB, CAN.
- **OS Concepts:** Cache Hierarchy, Scheduling, Process Management, Virtual Memory.
- **Tools:** Keil μVision, Proteus.
- **Debugging Tools:** JTAG, GDB.
- **AI Tools:** Claude, Cursor.
- **Version Control Tools:** Git/GitHub, VS Code.

Linux Kernel and Driver Development

- **Linux Device Drivers:** Strong knowledge of character device drivers (file operations, IOCTL handling, data transfer).
- **Platform Drivers:** GPIO, UART, I2C and Device Tree configuration/integration.
- **Interrupt Management:** Skilled in implementing interrupt handlers with bottom-half mechanisms like tasklets, workqueues, and kernel threads.
- **System Reliability:** Exposure to kernel crash dump analysis (Oops, panics) and identifying memory leaks.
- **Memory Handling:** Practical understanding of kernel memory allocation techniques including `kmalloc`, `vmalloc`.
- **USB 3.0:** Linux USB 3.0 host driver development including Device Descriptors, SuperSpeed Descriptors, Bulk-Only Transport (BOT), SCSI commands and Link Power Management (LPM).

Projects

USB3.0 Device Driver — Linux Kernel / USB

Role: Embedded Linux Kernel Developer.

- Developed a Linux USB 3.0 device driver to communicate with SuperSpeed USB devices and gain hands-on experience with the USB 3.x host-driver architecture.
- Implemented USB device communication and parsed Device, Configuration, Interface and Endpoint SuperSpeed Endpoint Companion Descriptors during device enumeration.
- Developed a strong understanding of the USB enumeration process, USB 3.0 bus architecture and various USB transfer types, including Control, Bulk, Interrupt and Isochronous transfers.
- Explored USB Link Power Management (LPM) and SuperSpeed power states to understand power-efficient USB device operation.
- Gained practical knowledge of xHCI ring buffers, Transfer Request Blocks (TRBs), USB packets, and host-controller data transfer mechanisms.

Tools & Technologies: C Programming, Linux Kernel, USB, SCSI, USB BOT, USB 3.x, xHCI, Linux Device Drivers.

Automated Temperature Monitoring and Controlling System — Embedded Linux

Role: Embedded Linux Kernel Developer.

Responsibilities & Contributions:

- Designed and developed a Linux kernel-space temperature monitoring and controlling system on Raspberry Pi and x86 platforms using custom character device drivers.

- Interfaced MCP3208, LM35 and LDR sensors using SPI and GPIO subsystems for real-time temperature and environmental monitoring.
- Used workqueues and wait queues to defer processing from interrupt context and synchronize user-space data access.
- Developed buffered sensor data handling and asynchronous event processing for efficient kernel-space communication.
- Performed low-level programming using `ioremap()`, `readl()` and `writel()`.
- Developed real-time threshold detection and alert generation for abnormal temperature conditions using kernel-space event handling.

Tools & Technologies: C Programming, Linux Device Drivers, UART, SPI, PWM, Kernel Threads, Raspberry Pi 4 Model B, MCP3208, LM35, x86 Systems.

Linux Shell Implementation Linux / C

- Implemented a custom shell supporting basic Linux commands, file operations, pipelining and signal handling.
- Built using Linux system calls, signal handling and file management while maintaining source code using Git.
- Gained hands-on experience with Linux internals and process management.

Tools & Technologies: C Programming, Git, Makefile, Linux System Programming.

Vehicle Dashboard Implementation using CAN Embedded C / CAN

- Designed and developed a CAN-based automotive dashboard system using LPC2129 for real-time vehicle parameter monitoring.
- Developed low-level CAN driver initialization and message handling using register-level programming.
- Integrated real-time display of vehicle parameters such as speed, temperature, fuel level and warning indicators.
- Gained hands-on experience in low-level firmware development, CAN protocol and error checking mechanisms.

Tools & Technologies: C Programming, Keil, Embedded-C, LED, 16x2 Display, Flash Magic.

Student Data Base Management System C / Data Structures

- Implemented a dynamic Student Database Management System with CRUD operations including Add, Update, Delete, Search, and Display.
- Applied linked lists, dynamic memory allocation, pointer manipulation, and file handling to manage student records through a menu-driven command-line interface.

Tools & Technologies: C Programming, Singly Linked List, Dynamic Memory Allocation, GCC, Makefile, Bash.

Academic Credentials

Bachelor's in Electrical and Electronics Engineering Aug 2024
CGPA: 8.43
Sri Venkateswara College of Engineering and Technology, Chittoor

Intermediate (MPC) Jun 2020
CGPA: 8.29
Mother Theresa Junior College, Palamaner

Secondary Education Mar 2018
CGPA: 9.2
Zillaparishad High School, Bandlapalli